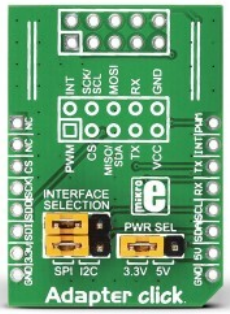
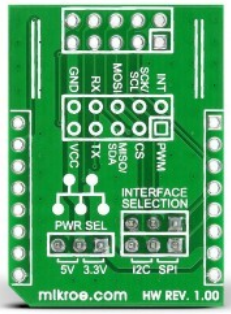
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Adapter Click





PID: MIKROE-1432

Adapter Click is an adapter board that simplifies the connection of add-ons with IDC10 headers to the mikroBUS™ socket. One 2x5-position 2.54mm pitch header makes an easy and elegant way to add the external connections to your development system, thorugh a mikroBUS™ socket. Each mikroBUS™ line is covered, except AN and RST, with two jumpers for communication selection and one for selection of power supply range.

Adapter Click is supported by a mikroSDK compliant library, which includes functions that simplify software development. This Click board™ comes as a fully tested product, ready to be used on a system equipped with the mikroBUS™ socket.

How does it work?

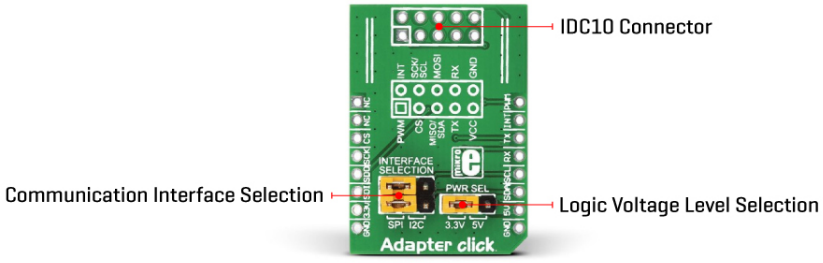
Adapter Click represents an adapter board that simplifies the connection of add-ons with IDC10 headers to the mikroBUS™ socket. One 2x5-position 2.54mm pitch header allows you to add sensors, drivers, and various components in a way that suits your project needs. Each header pin corresponds to a pin on the mikroBUS™ socket (except AN and RST pins). Thanks to these pins, the connection with the Click board™ remains firm and stable, always retaining a perfect connection quality.

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Committed to excellency, we are dedicated to helping engineers bring the project development up to speed and achieve outstanding results.

ISO 27001: 2013 certification of informational security management system.
ISO 14001: 2015 certification of environmental management system.
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There are two ways to establish such a connection: male or female IDC10 connectors. Both are provided with the package. You may solder the male IDC10 header on the top side of the Adapter Click and connect the add-on board directly or via IDC10 flat cable. In some cases, a female header socket is a better choice. Solder it either on the top or the bottom side, depending on which is more convenient in given circumstances.

Adapter Click allows using both I2C and SPI interfaces, where each mikroBUS™ line is covered, except, as mentioned before, AN and RST lines. The selection can be made by positioning jumpers labeled as INTERFACE SELECTION in an appropriate position marked as SPI or I2C. Note that all the jumpers' positions must be on the same side, or the Click board™ may become unresponsive.

This Click board™ can operate with both 3.3V and 5V logic voltage levels selected via the PWR SEL jumper. This way, it is allowed for both 3.3V and 5V capable MCUs to use the communication lines properly. However, the Click board™ comes equipped with a library containing easy-to-use functions and an example code that can be used, as a reference, for further development.

Specifications

Type	Adapter
Applications	Can be used to easily connect IDC10 add-ons to your developement boards with mikroBUS sockets
On-board modules	None
Key Features	Easy and elegant solution for adding the external connection capability to your development board
Interface	I2C,PWM,SPI,UART
Feature	No ClickID
Compatibility	mikroBUS™
Click board size	S (28.6 x 25.4 mm)
Input Voltage	3.3V or 5V

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Pinout diagram

This table shows how the pinout on Adapter Click corresponds to the pinout on the mikroBUS™ socket (the latter shown in the two middle columns).

Notes	Pin	mikroBUS				Pin	Notes
	NC	1	AN	PWM	16	**PWM**	PWM Signal
	NC	2	RST	INT	15	**INT**	Interrupt
SPI Chip Select	**CS**	3	CS	RX	14	**TX**	UART TX
SPI Clock	**SCK**	4	SCK	TX	13	**RX**	UART RX
SPI Data OUT	**SDO**	5	MISO	SCL	12	**SCL**	I2C Clock
SPI Data IN	**SDI**	6	MOSI	SDA	11	**SDA**	I2C Data
Power Supply	**3.3V**	7	3.3V	5V	10	**5V**	Power Supply
Ground	**GND**	8	GND	GND	9	**GND**	Ground

Onboard settings and indicators

Label	Name	Default	Description
JP1	PWR SEL	Left	Logic Level Voltage Selection 3V3/5V: Left position 3V3, Right position 5V
JP2	INTERFACE SELECTION	Left	Communication Interface Selection SPI/I2C: Left position SPI, Right position I2C

Adapter Click electrical specifications

Description	Min	Typ	Max	Unit
Supply Voltage	3.3	-	5	V

Software Support

We provide a library for the Adapter Click as well as a demo application (example), developed using MikroElektronika compilers. The demo can run on all the main MikroElektronika development boards.

Package can be downloaded/installed directly from NECTO Studio Package Manager(recommended way), downloaded from our LibStock™ or found on Mikroe github account.

Library Description

This library contains API for Adapter Click driver.

Key functions

- This function writes data to the desired register.
- This function reads data from the desired register.

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Example Description

Adapter click is a breakout board which simplifies connection of add-on boards. There are two ways of establishing connection: using male or female IDC10 connectors. Male header must be soldered on the top side of Adapter Click to connect the add-on board directly or via flat cable. Female header can be soldered either on the top, or the bottom side, depending on which one is more convenient in given circumstances. There are two jumpers for SPI/I2C selection and one for selection of power supply range.

The full application code, and ready to use projects can be installed directly from NECTO Studio Package Manager(recommended way), downloaded from our LibStock™ or found on Mikroe github account.

Other Mikroe Libraries used in the example:

- MikroSDK.Board
- MikroSDK.Log
- Click.Adapter

Additional notes and informations

Depending on the development board you are using, you may need USB UART click, USB UART 2 Click or RS232 Click to connect to your PC, for development systems with no UART to USB interface available on the board. UART terminal is available in all MikroElektronika compilers.

mikroSDK

This Click board™ is supported with mikroSDK - MikroElektronika Software Development Kit. To ensure proper operation of mikroSDK compliant Click board™ demo applications, mikroSDK should be downloaded from the LibStock and installed for the compiler you are using.

For more information about mikroSDK, visit the official page.

Resources

Click board™ Catalog

Click Boards™

Downloads

Adapter Click User Manual

Adapter click 2D and 3D files

Adapter click schematic

Adapter click example on Libstock

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